# Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

## STATEMENT

Graduate-level **AI Engineer** with 3 years of industry experience building and shipping working AI systems: **LLM applications**, **agentic tool-calling** workflows and **RAG** services for Compliance, Client Service and Investment teams in regulated financial services, taken from requirements through release and iteration. Applies governance and cost thinking in practice (department-level access control, LDAP **RBAC**, Compliance review, security-team **VAPT**, cost-aware model choice). Co-built a scoring and ranking tool with a **RAG** assistant for an MSc dissertation with AstraZeneca and placed 2nd at the Encode hackathon with a **spec-driven** build. De facto team lead for up to 4 interns; twice recognised with Best Team Project.

## EDUCATION

**University of Bristol – MSc. Data Science (awaiting result)** September 2025 – September 2026
**NMIMS B.Tech Information Technology (Dual Degree)** June 2017 – April 2022
**NMIMS MBA Technology Management (Dual Degree)** June 2017 – April 2022

## WORK EXPERIENCE

**HDFC Asset Management Company Ltd.** July 2022 – July 2025

**AI Solutions Engineer / Forward Deployed Engineer**

### AI Engineering, Agentic Systems & Governance

- Specified, designed and built **LLM applications** end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & Llama-3) for Compliance, Client Service and Investment, with **Azure OpenAI** embeddings and a separate **pgvector** database per department so each team searched only its own internal documents.
- Built **tool-calling** agents that call **Python** functions as tools to run **SQL**-backed fund comparisons and generate PDF reports on request; guided behaviour with system prompts, versioned every prompt change in **Git**, released to real users, changed the applications based on what they found, and had the Compliance department review each release.
- Weighed cost against quality when choosing models: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models as needs changed.

### Production Delivery, Security & Reliability

- Turned prototypes into deployable tools: packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (**EC2**, **ECR**) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.
- Coordinated with the security team on which ports to whitelist and which safeguards to maintain.

### Requirements, Data & Stakeholder Delivery

- Worked directly with business stakeholders in Client Service, Compliance and Investment to turn requests into clear requirements, demonstrate tools and turn feedback into releases; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

- Built a **Python** extraction pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that parses over **300** daily emails from credit rating agencies into structured data, fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the Morningstar API covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy analyst data; developed **Optimized SQL** queries, Functions and Procedures.
- Analyzed customer data using **Python Pandas** and **Market Basket Analysis** (**Apriori Algorithm**) to identify buying patterns, handing the insights to the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10** hours a week.

## PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring algorithm and RAG-based AI assistant for a tool, built as FastAPI microservices with a React frontend, that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick, using per-gene scores across six evidence layers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes, stored as data so tuning a persona needs no code change, and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders with symbolic, SBERT embedding-retrieval and LSTM sequence decoders (with and without Bahdanau attention), trained over five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

## ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

## SKILLS

- **Tools:** Claude Code (AI-assisted development), codeplain (spec-driven development), Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite